Ladder Operators

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1 Harmonic Oscillator

The purpose of this note is to examine some elementary applications of quantum mechanics. Another purpose of this note is to look at how the intrinsic operator algebra of quantum mechanics can easily solve some of the Schrödinger equations that may be difficult to solve in its differential equation forms.

The quantum harmonic oscillator is ubiquitous in modern day physics and chemistry. In chemistry, it is useful as a zeroth order model for the vibration of molecules. In physics, the model is often used in a more abstract sense by treating particles as excitation of fields.

We wish to solve the TISE of the system, which has stationary states and energy spectrum as solutions

$$H|n\rangle = E_n|n\rangle \tag{1.1}$$

The Hamiltonian is given by

$$H = \frac{p^2}{2m} + \frac{1}{2}kx^2 \tag{1.2}$$

which can be rearranged into the form

$$H = \frac{(m\omega x)^2 + p^2}{2m} \tag{1.3}$$

The fundamental commutator for this problem is the canonical commutation relation

$$[x, p] = i\hbar \tag{1.4}$$

We now introduce the **ladder operators** by factorising the Hamiltonian

$$\boxed{a \equiv \frac{m\omega x + ip}{\sqrt{2m\hbar\omega}}} \tag{1.5}$$

$$\boxed{a^\dagger = \frac{m\omega x - ip}{\sqrt{2m\hbar\omega}}} \tag{1.6}$$

which combines to give

$$aa^\dagger = \frac{H}{\hbar\omega} + \frac{1}{2} \tag{1.7}$$

$$a^\dagger a = \frac{H}{\hbar\omega} - \frac{1}{2} \tag{1.8}$$

$a^\dagger$ is known as the **creation operator** and a is known as the **annihilation operator**. consider

$$\begin{aligned} Ha|n\rangle &= \hbar\omega\Big(aa^\dagger - \frac{1}{2}\Big)a|n\rangle \\ &= \hbar\omega a\Big(a^\dagger a - \frac{1}{2}\Big)|n\rangle \\ &= a(H - \hbar\omega)|n\rangle \\ &= (E_n - \hbar\omega)a|n\rangle \end{aligned} \tag{1.9}$$

Thus $a|n\rangle$ is another eigenstate of H, with a different eigenvalue

$$a|n\rangle \propto |n-1\rangle \tag{1.10}$$

$$E_{n-1} = E_n - \hbar\omega \tag{1.11}$$

Repeating the calculation by considering $Ha^\dagger|n\rangle$ yields

$$a^\dagger|n\rangle \propto |n+1\rangle \tag{1.12}$$

$$E_{n+1} = E_n + \hbar\omega \tag{1.13}$$

Thus we have a "ladder" of energy eigenvalues, adjacent energy levels are separated by $\hbar\omega$. We now consider whether it is possible for the ladder to continue infinitely.

There is no reason that there is an upper limit of the energy. However, there is a lower limit, because H is a sum of Hermitian operators x and p, which must has non-negative eigenvalues

$$\begin{aligned} E_n &= \langle n|H|n\rangle \\ &= \frac{\langle n|p^2|n\rangle}{2m} + \frac{1}{2}k\langle n|x^2|n\rangle \\ &= \frac{|p|n\rangle|^2}{2m} + \frac{1}{2}k|x|n\rangle|^2 \\ &\geq 0 \end{aligned} \tag{1.14}$$

This suggests there exist a ground state $|0\rangle$ which satisfies the following relation

$$a|0\rangle = 0 \tag{1.15}$$

Thus we can calculate the **zero-point energy** of the harmonic oscillator

$$\begin{aligned} E_0 &= \langle 0|H|0\rangle \\ &= \hbar\omega\langle 0|a^\dagger a + \frac{1}{2}|0\rangle \\ &= \hbar\omega\langle 0|a^\dagger a|0\rangle + \frac{1}{2}\hbar\omega\langle 0|0\rangle \end{aligned}$$

$$E_0 = \frac{1}{2}\hbar\omega \tag{1.16}$$

where we have also used $\langle 0|0\rangle = 1$.

Using (1.13), we have the rest of the eigenvalues

$$\boxed{E_n = \left(n + \frac{1}{2}\right)\hbar\omega} \tag{1.17}$$

We shall now normalise the eigenstates

$$
\begin{aligned}
|n+1\rangle &= \frac{a^\dagger|n\rangle}{\sqrt{\langle n|aa^\dagger|n\rangle}} \\
&= \frac{a^\dagger|n\rangle}{\sqrt{\langle n|H/\hbar\omega + 1/2|n\rangle}} \\
&= \frac{a^\dagger|n\rangle}{\sqrt{E_n/\hbar\omega + 1/2}}
\end{aligned}
$$

$$
\boxed{|n+1\rangle = \frac{a^\dagger|n\rangle}{\sqrt{n+1}}} \tag{1.18}
$$

Similarly

$$
\boxed{|n-1\rangle = \frac{a|n\rangle}{\sqrt{n}}} \tag{1.19}
$$

Furthermore, repeated substitution of recurrence relation (1.18) into itself yields

$$
|n\rangle = \frac{(a^\dagger)^n|0\rangle}{\sqrt{n!}} \tag{1.20}
$$

We shall now calculation the wavefunctions, begin by writing (1.15) in position representation, then rearrange

$$
\Big(\frac{m\omega x}{\hbar} + \frac{d}{dx}\Big)\langle x|0\rangle = 0 \tag{1.21}
$$

This is a linear first order ODE, which can be solved by seperation of variables, the solution is

$$
\boxed{\langle x|0\rangle \propto e^{-m\omega x^2/2\hbar}} \tag{1.22}
$$

Wavefunction can be normalised by setting the integral $\int dx\, \langle 0|x\rangle\langle x|0\rangle$ to 1.

Wavefunctions of the excited states can be obtained by applying $a^\dagger$ in position representation using (1.20).

2 Angular Momentum

The theory of angular momentum is enormously important in quantum mechanics. First of all, consideration of angular momentum is crucial for any systems with circular and spherical symmetry, which include atoms. Furthermore, we will see shortly that angular momentum is quantised, which has far reaching consequences such as shaping the energy levels of atoms. It also governs most of the selection rules, which is important for spectroscopy. There are various types of angular momentum and they can couple in various ways, the coupling does not require energy and it is inherently quantum mechanical. One of the consequences of coupling of angular momentum is that the two sources of angular momentum becomes statistically correlated, which leads to the phenomenon of **quantum entanglement**. The angular momentum eigenvalue problem is also the **rigid rotor** model (up to a constant factor), which is used as a zeroth order model for molecular rotation. Lastly, we shall see that spin has the algebraic structure of angular momentum and the construction of many important modern technologies based on spin relies on the theory of angular momentum.

We begin with the commutation relations of angular momentum

$$[J^2, J_i] = 0 \tag{2.1}$$

$$[J_i, J_j] = i\hbar \sum_k \epsilon_{ijk} J_k \tag{2.2}$$

Commutation relation (2.1) implies there exist a complete set of mutual eigenstates $\{|j, m\rangle\}$, which safisfies the following eigenvalue equations

$$J^2|j, m\rangle = f_j \hbar^2 |j, m\rangle \tag{2.3}$$

$$J_z|j, m\rangle = m\hbar|j, m\rangle \tag{2.4}$$

We we have noticed $\hbar$ has the unit of angular momentum, thus is used to carry the unit of the eigenvalue. (2.2) suggests we can only specify one component of angular momentum along with the total angular momentum, we have chosen the z direction by convention. Furthermore, we have used j to label the eigenvalue of J^2 for future convenience, which shall be apparent shortly. The reason is the same as using n to label the eigenvalue of energy in (1.1).

It is useful to write down the expression

$$J^2 = J_x^2 + J_y^2 + J_z^2 \tag{2.5}$$

Similar to the analysis of harmonic oscillator, we introduce the ladder operators by factorising the total angular momentum (2.5)

$$\boxed{J_\pm \equiv J_x \pm i J_y} \tag{2.6}$$

which combines to give

$$J_\pm J_\mp = J^2 - J_z^2 \pm \hbar J_z \tag{2.7}$$

J_+ is known as **raising operator** and J_- is known as **lowering operator**.

It is also worth noting $J_\pm^\dagger = J_\mp$.
The ladder operators have the following commutation relations

$$[J^2, J_\pm] = 0 \tag{2.8}$$

$$\begin{aligned}
[J_z, J_\pm] &= [J_z, J_x \pm iJ_y] \\
&= [J_z, J_x] \pm i[J_z, J_y] \\
&= +i\hbar J_y \pm \hbar J_x \\
&= \pm\hbar(J_x \pm iJ_y) \\
&= \pm\hbar J_\pm
\end{aligned} \tag{2.9}$$

Consider

$$\begin{aligned}
J^2 J_\pm |j,m\rangle &= J_\pm J^2 |j,m\rangle \\
&= f_j \hbar^2 J_\pm |j,m\rangle
\end{aligned} \tag{2.10}$$

$$\begin{aligned}
J_z J_\pm |j,m\rangle &= (J_\pm J_z + [J_z, J_\pm])|j,m\rangle \\
&= (m\hbar J_\pm \pm \hbar J_\pm)|j,m\rangle \\
&= (m \pm 1)\hbar J_\pm |j,m\rangle
\end{aligned} \tag{2.11}$$

Thus $J_\pm|j,m\rangle$ is another mutual eigenstates of J^2 and J_z

$$J_\pm |j,m\rangle \propto |j, m \pm 1\rangle \tag{2.12}$$

J^2 is a square of a Hermitian operators, thus it must have non-negative eigenvalue, furthermore

$$\begin{aligned}
f_j\hbar^2 &= \langle j,m|J^2|j,m\rangle \\
&= \langle j,m|J_x^2 + J_y^2 + J_z^2|j,m\rangle \\
&= \langle j,m|J_x^2|j,m\rangle + \langle j,m|J_y^2|j,m\rangle + \langle j,m|J_z^2|j,m\rangle \\
&\geq m^2\hbar^2
\end{aligned} \tag{2.13}$$

Thus the ladder of J_z must terminate both at the top and bottom. We shall now define

$$j \equiv m_{max} \tag{2.14}$$

$$J_+|j,j\rangle = 0 \tag{2.15}$$

We can now extract the eigenvalues of J^2 using (2.7) and (2.15)

$$\begin{aligned}
f_j\hbar^2 &= \langle j,j|J^2|j,j\rangle \\
&= \langle j,j|J_z^2 + \hbar J_z + J_- J_+|j,j\rangle \\
&= j(j+1)\hbar^2
\end{aligned} \tag{2.16}$$

$$\boxed{f_j = j(j+1)} \tag{2.17}$$

where we have used $\langle j, j|j, j\rangle = 1$. Thus j does indeed conveniently label the total angular momentum.

Similarly

$$J_-|j, m_{min}\rangle = 0 \tag{2.18}$$

$$f_j = m_{min}(m_{min} - 1) \tag{2.19}$$

equate (2.19) to (2.17) then solve for m_{min} yields 2 solutions, the first solution is $m_{min} = j + 1$, which is not valid due to $j \geq m_{min}$. The second solution is the only valid solution

$$m_{min} = -j \tag{2.20}$$

From the ladder of eigenvalues of J_z, we observe there is an integer number of steps between $-j$ and j, thus j is either an integer or half-integer and $m = -j, -j+1, \ldots, j$.

We shall now normalise the eigenstates

$$\begin{aligned} |j, m \pm 1\rangle &= \frac{J_\pm|j, m\rangle}{\sqrt{\langle j, m|J_\mp J_\pm|j, m\rangle}} \\ &= \frac{J_\pm|j, m\rangle}{\sqrt{\langle j, m|J^2 - J_z^2 \mp \hbar J_z|j, m\rangle}} \\ &= \frac{J_\pm|j, m\rangle}{\sqrt{j(j+1)\hbar^2 - m^2\hbar^2 \mp m\hbar^2}} \end{aligned} \tag{2.21}$$

$$\boxed{|j, m \pm 1\rangle = \frac{J_\pm|j, m\rangle}{\hbar\sqrt{j(j+1) - m(m \pm 1)}}} \tag{2.22}$$

The analysis above is valid for any operators that satisfy commutation relation (2.1) and (2.2). We shall now consider **orbital angular momentum**, which is defined in the same way as the classical analogue

$$\mathbf{L} \equiv \mathbf{x} \times \mathbf{p} \tag{2.23}$$

which can be shown to satisfy (2.1) and (2.2) but with J_i replaced by L_i

$$[L^2, L_i] = 0 \tag{2.24}$$

$$[L_i, L_j] = i\hbar \sum_k \epsilon_{ijk} L_k \tag{2.25}$$

Now consider a small translation on a circle with $\mathbf{n}$ as the normal vector, and distance translated is $\delta\mathbf{a} = \delta a\, \mathbf{n} \times \mathbf{x}$

$$\begin{aligned} T(\delta\mathbf{a}) &= e^{-i\delta\mathbf{a}\cdot\mathbf{p}/\hbar} \\ &= e^{-i\delta a\,(\mathbf{n}\times\mathbf{x})\cdot\mathbf{p}/\hbar} \\ &= e^{-i\delta a\,\mathbf{n}\cdot(\mathbf{x}\times\mathbf{p})/\hbar} \\ &= e^{-i\delta a\,\mathbf{n}\cdot\mathbf{L}/\hbar} \end{aligned} \tag{2.26}$$

for large translations on a circle, we can see it as a lot of successive small translations.

$$R(\boldsymbol{\alpha}) = e^{-i\boldsymbol{\alpha}\cdot\mathbf{L}/\hbar} \tag{2.27}$$

where we have defined $\boldsymbol{\alpha} \equiv \alpha\mathbf{n}$.
Thus, $\mathbf{L}$ is the generator of translation on a circle
Furthermore, we have the commutation relation

$$[\phi, L_z] = i\hbar \tag{2.28}$$

hence

$$[\phi, R(\alpha)] = \alpha R(\alpha) \tag{2.29}$$

where $\phi \equiv \arctan(y/x)$, note ϕ is a Hermitian operator with eigenvalue equation

$$\phi|\phi'\rangle = \phi'|\phi'\rangle \tag{2.30}$$

Consider

$$\begin{aligned} \phi R(\alpha)|\phi'\rangle &= (R(\alpha)\phi + [\phi, R(\alpha)])|\phi'\rangle \\ &= (R(\alpha)\phi + \alpha R(\alpha))|\phi'\rangle \\ &= (\phi' + \alpha)R(\alpha)|\phi\rangle \end{aligned} \tag{2.31}$$

thus $R(\alpha)|\phi'\rangle$ is another eigenstate and $R(\alpha)$ is contineous spectrum analogue of ladder operators discussed previously

$$R(\alpha)|\phi'\rangle = |\phi' + \alpha\rangle \tag{2.32}$$

Note the normalisation constant is 1 as $R(\alpha)$ is unitary, thus does not change the length of the ket.

Now consider translation on a closed path, back to the same starting point with no net displacement, thus $\sum_i \mathbf{a}_i = \mathbf{0}$

$$\begin{aligned} \prod_i T(\mathbf{a}_i) &= \prod_i e^{-i\mathbf{a}_i\cdot\mathbf{p}/\hbar} \\ &= e^{-i\sum_i \mathbf{a}_i\cdot\mathbf{p}/\hbar} \\ &= e^{-i\mathbf{0}\cdot\mathbf{p}/\hbar} \\ &= I \end{aligned} \tag{2.33}$$

A circular path is necessarily closed, thus $R(2\pi) = I$ and

$$\begin{aligned}|\ell, m_\ell\rangle &= R(2\pi)|\ell, m_\ell\rangle \\ &= e^{-2\pi i L_z/\hbar}|\ell, m_\ell\rangle \\ &= e^{-2\pi m_\ell i}|\ell, m_\ell\rangle \end{aligned} \tag{2.34}$$

hence

$$e^{-2\pi m_\ell i} = 1 \tag{2.35}$$

thus m_ℓ must be an integer, which implies ℓ must also be an integer.

We shall now calculate the wavefunctions of orbital angular momentum eigenstates

$$Y_\ell^m(\theta, \phi) \equiv \langle \theta, \phi | \ell, m_\ell \rangle \tag{2.36}$$

which are more commonly known as **spherical harmonics**

The total angular momentum, angular momentum in z-direction and ladder operators in polar coordinates of position representation are as follows

$$L^2 = -\hbar^2 \left(\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \Big(\sin\theta \frac{\partial}{\partial\theta} \Big) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\phi^2} \right) \tag{2.37}$$

$$L_z = -i\hbar \frac{\partial}{\partial\phi} \tag{2.38}$$

$$L_\pm = \hbar e^{\pm i\phi} \left(\pm \frac{\partial}{\partial\theta} + i \cot\theta \frac{\partial}{\partial\phi} \right) \tag{2.39}$$

Thus, the eigenvalue equation $L_z|\ell, m_\ell\rangle = m_\ell \hbar |\ell, m_\ell\rangle$ in position representation is

$$-i\hbar \frac{\partial}{\partial\phi} Y_\ell^m(\theta, \phi) = m_\ell \hbar Y_\ell^m(\theta, \phi) \tag{2.40}$$

This is a linear first order ODE, which can be solved by separation of variables, the solution is the ϕ dependence of the wavefunction

$$Y_\ell^m(\theta, \phi) \propto \Theta_\ell^m(\theta) e^{i m_\ell \phi} \tag{2.41}$$

to determine $\Theta_\ell^m(\theta)$, we begin by writing $L_+|\ell, \ell\rangle = 0$ in position representation

$$\hbar e^{i\phi} \left(\frac{\partial}{\partial\theta} + i \cot\theta \frac{\partial}{\partial\phi} \right) Y_\ell^\ell(\theta, \phi) = 0 \tag{2.42}$$

Substitution of (2.41) into (2.42) then rearrange yields

$$\Big(\frac{\partial}{\partial\theta} - \ell \cot\theta \Big) \Theta_\ell^\ell(\theta) = 0 \tag{2.43}$$

this again can be solved by separation of variables, the solution is

$$\Theta_\ell^\ell(\theta) \propto \sin^\ell\theta \tag{2.44}$$

combine with (2.41) yields

$$\boxed{Y_\ell^\ell(\theta,\phi) \propto \sin^\ell\theta e^{i\ell\phi}} \tag{2.45}$$

similarly, by considering $L_-|\ell,-\ell\rangle = 0$, we obtain

$$\boxed{Y_\ell^{-\ell}(\theta,\phi) \propto \sin^\ell\theta e^{-i\ell\phi}} \tag{2.46}$$

every other spherical harmonics can be generated by using the ladder operators in position representation.

Lastly, we will look at the property spin angular momentum or simply **spin**.

Spin is defined as the difference

$$\mathbf{S} \equiv \mathbf{J} - \mathbf{L} \tag{2.47}$$

where $\mathbf{J}$ is the **total angular momentum**. This definition leads to the commutation relations

$$[S^2, S_i] = 0 \tag{2.48}$$

$$[S_i, S_j] = i\hbar \sum_k \epsilon_{ijk} S_k \tag{2.49}$$

thus the spectrum of spin immediately follows from the previous analysis on angular momentum from (2.3) to (2.22) by the replacement $J \to S$. However, unlike orbital angular momentum, m_s does not have to be integer and there is no position representation for the spin operators thus spin eigenstates. For example the spin of proton and electron is $1/2$, thus it has 2 eigenstates of S_z, conventionally denoted as

$$|\alpha\rangle \equiv |\frac{1}{2}, +\frac{1}{2}\rangle \tag{2.50}$$

$$|\beta\rangle \equiv |\frac{1}{2}, -\frac{1}{2}\rangle \tag{2.51}$$

3 Hydrogen Atom

Quantum mechanics was originally developed as an attempt to understand the structure of atoms. Hydrogen contains one proton and one electron, thus it is the simplest possible atom. Furthermore, hydrogen atom is used as a zeroth order model to the electronic structure of other atoms and molecules, and the periodic table is constructed based on the stationary states of the hydrogen atom. The study of the structure of hydrogen atom is a starting point of rest of atomic physics and chemistry.

Ignoring spin and relativistic effects, Hamiltonian of a hydrogenic atom is written as

$$H = \frac{p_N^2}{2m_N} + \frac{p_e^2}{2m_e} - \frac{Ze^2}{|\mathbf{x}_e - \mathbf{x}_N|} \tag{3.1}$$

this Hamiltonian describes the **gross structure** of hydrogenic atoms.

By exploiting the spherical symmetry of the potential, the Hamiltonian can be rewritten as

$$H = \frac{p_r^2}{2\mu} + \frac{L^2}{2\mu r^2} - \frac{Z}{4\pi\epsilon_0 r} \tag{3.2}$$

where $\mu = m_e m_N/(m_e + m_N)$ is the reduced mass, which is very close to electron mass m_e for any hydrogenic atom. m_N is nuclear mass.

radial position operator is defined as

$$r \equiv \sqrt{x^2 + y^2 + z^2} \tag{3.3}$$

and **radial momentum operator** is defined as

$$p_r \equiv \frac{1}{2}\Big(\frac{\mathbf{r}}{|\mathbf{r}|} \cdot \mathbf{p} + \mathbf{p} \cdot \frac{\mathbf{r}}{|\mathbf{r}|}\Big) \tag{3.4}$$

The fundamental commutator for this problem is

$$[r, p_r] = i\hbar \tag{3.5}$$

As H, L^2 and L_z all mutually commute, there exist a complete set of mutual eigenstates $\{|n, \ell, m_\ell\rangle\}$, which can be factoriesd as

$$|n, \ell, m_\ell\rangle = |n, \ell\rangle|\ell, m_\ell\rangle \tag{3.6}$$

where we have already solved for $|\ell, m_\ell\rangle$ in the previous chapter.

Note H acting on the a mutual eigenstate has the same effect as the radial Hamiltonian defined as

$$H_\ell \equiv \frac{p_r^2}{2\mu} + \frac{\ell(\ell+1)\hbar^2}{2\mu r^2} - \frac{Z}{4\pi\epsilon_0 r} \tag{3.7}$$

which can be rearranged into a more convenient form for factorisation

$$H_\ell = \frac{\hbar^2}{2\mu}\Big(\frac{p_r^2}{\hbar^2} + \frac{\ell+1}{r^2} - \frac{2Z}{ra_0}\Big) \tag{3.8}$$

where have defined **Bohr radius**

$$a_0 \equiv \frac{4\pi\epsilon_0\hbar^2}{\mu e^2} \approx 0.529\text{Å} \tag{3.9}$$

The eigenvalue equation we wish to solve is the radial Schrödinger equation

$$H_\ell|n,\ell\rangle = E_n|n,\ell\rangle \tag{3.10}$$

Consider the following ladder operators

$$\boxed{C_\ell \equiv \frac{a_0}{\sqrt{2}}\Big(\frac{i}{\hbar}p_r - \frac{\ell+1}{r} + \frac{Z}{(\ell+1)a_0}\Big)} \tag{3.11}$$

$$\boxed{C_\ell^\dagger = \frac{a_0}{\sqrt{2}}\Big(-\frac{i}{\hbar}p_r - \frac{\ell+1}{r} + \frac{Z}{(\ell+1)a_0}\Big)} \tag{3.12}$$

which combines to give

$$C_\ell^\dagger C_\ell = \frac{a_0^2\mu}{\hbar^2}H_\ell + \frac{Z^2}{2(\ell+1)^2} \tag{3.13}$$

$$C_\ell C_\ell^\dagger = \frac{a_0^2\mu}{\hbar^2}H_{\ell+1} + \frac{Z^2}{2(\ell+1)^2} \tag{3.14}$$

We now demonstrate the proposed operators are indeed ladder operators

$$\begin{aligned} H_{\ell+1}C_\ell|n,\ell\rangle &= \frac{\hbar^2}{a_0^2\mu}\Big(C_\ell C_\ell^\dagger - \frac{Z^2}{2(\ell+1)^2}\Big)C_\ell|n,\ell\rangle \\ &= \frac{\hbar^2}{a_0^2\mu}C_\ell\Big(C_\ell^\dagger C_\ell - \frac{Z^2}{2(\ell+1)^2}\Big)|n,\ell\rangle \\ &= C_\ell H_\ell|n,\ell\rangle \\ &= E_n C_\ell|n,\ell\rangle \end{aligned} \tag{3.15}$$

thus we have generated another eigenstate

$$C_\ell|n,\ell\rangle \propto |n,\ell+1\rangle \tag{3.16}$$

similarly

$$C_\ell^\dagger|n,\ell+1\rangle \propto |n,\ell\rangle \tag{3.17}$$

A closer examination of the effective potential

$$V_{eff}(r) \equiv \frac{\ell(\ell+1)\hbar^2}{2\mu r^2} - \frac{Ze^2}{4\pi\epsilon_0 r} \tag{3.18}$$

shows the potential well becomes shallower as ℓ increases, therefore the ladder must terminate, thus there exist a state with $\lambda \equiv \ell_{max}$ which satisfies the following relation

$$C_\lambda|n,\lambda\rangle = 0 \tag{3.19}$$

the energy spectrum can then be deduced

$$\begin{aligned} E_n &= \langle n,\lambda|H_\lambda|n,\lambda\rangle \\ &= \frac{\hbar^2}{a_0^2\mu}\langle n,\lambda|C_\lambda^\dagger C_\lambda - \frac{Z^2}{2(\ell+1)^2}|n,\lambda\rangle \\ &= -\frac{\hbar^2 Z^2}{2a_0^2\mu(\lambda+1)^2} \end{aligned} \tag{3.20}$$

Defining the **principal quantum number**

$$n \equiv \lambda + 1 \tag{3.21}$$

and **Rydberg constant**

$$R \equiv \frac{\hbar^2}{2a_0^2\mu} = \frac{e^2}{8\pi\epsilon_0 a_0} \approx 13.6 \text{ eV} \tag{3.22}$$

the energy eigenvalue thus can be rewritten as

$$\boxed{E_n = -R\frac{Z^2}{n^2}} \tag{3.23}$$

We shall now normalise the eigenstates

$$\begin{aligned} |n,\ell+1\rangle &= \frac{C_\ell|n,\ell\rangle}{\sqrt{\langle n,\ell|C_\ell^\dagger C_\ell|n,\ell\rangle}} \\ &= \frac{C_\ell|n,\ell\rangle}{\sqrt{\langle n,\ell|\frac{a_0^2\mu}{\hbar^2}H_\ell + \frac{Z^2}{2(\ell+1)^2}|n,\ell\rangle}} \\ &= \frac{C_\ell|n,\ell\rangle}{\sqrt{\frac{a_0^2\mu}{\hbar^2}E_n + \frac{Z^2}{2(\ell+1)^2}}} \\ &= \frac{C_\ell|n,\ell\rangle}{\sqrt{-\frac{Z^2}{2n^2} + \frac{Z^2}{2(\ell+1)^2}}} \end{aligned} \tag{3.24}$$

$$\boxed{|n,\ell+1\rangle = \frac{C_\ell|n,\ell\rangle}{Z\sqrt{2}^{-1}\sqrt{(\ell+1)^{-2} - n^{-2}}}} \tag{3.25}$$

similarly

$$\boxed{|n,\ell-1\rangle = \frac{C_{\ell-1}^\dagger|n,\ell\rangle}{Z\sqrt{2}^{-1}\sqrt{\ell^{-2} - n^{-2}}}} \tag{3.26}$$

We shall now calculate the radial wavefunctions $R_{n\ell}(r) \equiv \langle r|n,\ell\rangle$, begin by casting (3.19) into the position representation

$$\Big(\frac{\partial}{\partial r} - \frac{n-1}{r} + \frac{Z}{na_0}\Big)R_{n,n-1}(r) = 0 \tag{3.27}$$

where we have used

$$p_r = -i\hbar\Big(\frac{\partial}{\partial r} + \frac{1}{r}\Big) \tag{3.28}$$

(3.27) is a linear first order ODE, and can be solved simply by separation of variables, the solution is

$$\boxed{R_{n,n-1}(r) \propto r^{n-1}e^{-Zr/na_0}} \tag{3.29}$$

these are the radial wavefunctions of $1s, 2p, 3d$ and $4f$ **atomic orbitals**, $2s, 3p$ and $4d$ orbitals can be obtained by applying $C^\dagger_{n-2}$ in position representation to $R_{n,n-1}$, and so on. Note it is conventional to use letters to label ℓ, where $\ell = 0, 1, 2, 3$ corresponds to s, p, d, f respectively, these labels originates from spectroscopy.

The full wavefunction is a product of the radial wavefunction and spherical harmonics discussed in the previous chapter

$$\psi^m_{n\ell}(r, \theta, \phi) = R_{n\ell}(r)Y^m_\ell(\theta, \phi) \tag{3.30}$$

which is just (3.6) written in position representation

4 Isotropic Harmonic Oscillator

We have solved 1D harmonic oscillator in section one. However, the world is 3 dimensional, thus it is more physically realistic to look at a 3D isotropic oscillator, which has spherical symmetry and a richer dynamics than the 1D analogue.

The Hamiltonian for an isotropic harmonic oscillator is written as

$$H = \frac{p_r^2}{2\mu} + \frac{L^2}{2\mu r^2} + \frac{1}{2}\mu\omega^2 r^2 \tag{4.1}$$

The fundamental commutator in this problem is analogous to the canonical commutation relation

$$[r, p_r] = i\hbar \tag{4.2}$$

Similar to Hydrogen, H, L^2 and L_z all mutually commute, there exist a complete set of mutual eigenstates $\{|n, \ell, m_\ell\rangle\}$, which can be factorised as

$$|n, \ell, m_\ell\rangle = |n, \ell\rangle|\ell, m_\ell\rangle \tag{4.3}$$

Note H acting on the a mutual eigenstate has the same effect as the radial Hamiltonian defined as

$$H_\ell \equiv \frac{p_r^2}{2\mu} + \frac{\ell(\ell+1)\hbar^2}{2\mu r^2} + \frac{1}{2}\mu\omega^2 r^2 \tag{4.4}$$

The radial TISE is written as

$$H_\ell|n, \ell\rangle = E_n|n, \ell\rangle \tag{4.5}$$

Consider the following 2 sets of operators

$$\boxed{\zeta_\ell \equiv \frac{1}{\sqrt{2\mu\hbar\omega}}\Big(ip_r - \frac{(\ell+1)\hbar}{r} + \mu\omega r\Big)} \tag{4.6}$$

$$\boxed{\zeta_\ell^\dagger = \frac{1}{\sqrt{2\mu\hbar\omega}}\Big(-ip_r - \frac{(\ell+1)\hbar}{r} + \mu\omega r\Big)} \tag{4.7}$$

$$\boxed{\xi_\ell \equiv \frac{1}{\sqrt{2\mu\hbar\omega}}\Big(ip_r + \frac{(\ell+1)\hbar}{r} + \mu\omega r\Big)} \tag{4.8}$$

$$\boxed{\xi_\ell^\dagger = \frac{1}{\sqrt{2\mu\hbar\omega}}\Big(-ip_r + \frac{(\ell+1)\hbar}{r} + \mu\omega r\Big)} \tag{4.9}$$

Which combines to give

$$\zeta_\ell\zeta_\ell^\dagger = \frac{H_{\ell+1}}{\hbar\omega} - \ell - \frac{1}{2} \tag{4.10}$$

$$\zeta_\ell^\dagger\zeta_\ell = \frac{H_\ell}{\hbar\omega} - \ell - \frac{3}{2} \tag{4.11}$$

$$\xi_\ell \xi_\ell^\dagger = \frac{H_\ell}{\hbar\omega} + \ell + \frac{3}{2} \tag{4.12}$$

$$\xi_\ell^\dagger \xi_\ell = \frac{H_{\ell+1}}{\hbar\omega} + \ell + \frac{1}{2} \tag{4.13}$$

We now demonstrate these are ladder operators, consider

$$\begin{aligned} H_{\ell+1}\zeta_\ell|n,\ell\rangle &= \hbar\omega\Big(\zeta_\ell\zeta_\ell^\dagger + \ell + \frac{1}{2}\Big)\zeta_\ell|n,\ell\rangle \\ &= \hbar\omega\zeta_\ell\Big(\zeta_\ell^\dagger\zeta_\ell + \ell + \frac{1}{2}\Big)|n,\ell\rangle \\ &= \zeta_\ell(H_\ell - \hbar\omega)|n,\ell\rangle \\ &= (E_n - \hbar\omega)\zeta_\ell|n,\ell\rangle \end{aligned} \tag{4.14}$$

Comparison with (4.5) yields

$$\zeta_\ell|n,\ell\rangle \propto |n-1,\ell+1\rangle \tag{4.15}$$

where

$$E_{n-1} = E_n - \hbar\omega \tag{4.16}$$

repeat the argument for other three operators yields

$$\zeta_\ell^\dagger|n,\ell+1\rangle \propto |n+1,\ell\rangle \tag{4.17}$$

$$\xi_\ell|n,\ell+1\rangle \propto |n-1,\ell\rangle \tag{4.18}$$

$$\xi_\ell^\dagger|n,\ell\rangle \propto |n+1,\ell+1\rangle \tag{4.19}$$

H is a sum of the squares of Hermitian operators, thus its eigenvalue must be non-negative, the ladder must terminate. Furthermore, analogous to hydrogen, there exist a state of maximum angular momentum for a state of well-defined energy, and we define

$$n \equiv \ell_{max} \tag{4.20}$$

Thus the ladder terminates in the following way

$$\zeta_n|n,n\rangle = 0 \tag{4.21}$$

We can then extract the eigenvalues

$$\begin{aligned} E_n &= \langle n,n|H_n|n,n\rangle \\ &= \hbar\omega\langle n,n|\zeta_n^\dagger\zeta_n + n + \frac{3}{2}|n,n\rangle \end{aligned} \tag{4.22}$$

$$\boxed{E_n = \Big(n + \frac{3}{2}\Big)\hbar\omega} \tag{4.23}$$

where we have used $\langle n', \ell' | n, \ell \rangle = \delta_{n'n}\delta_{\ell'\ell}$. Note the zero-point energy is $\hbar\omega/2$ in each dimension.

Newly generated ket can be normalised by dividing through by the square root of self inner product

$$\begin{aligned} |n-1, \ell+1\rangle &= \frac{\zeta_\ell |n, \ell\rangle}{\sqrt{\langle n, \ell | \zeta_\ell^\dagger \zeta_\ell | n, \ell \rangle}} \\ &= \frac{\zeta_\ell |n, \ell\rangle}{\sqrt{\langle n, \ell | H_\ell/\hbar\omega - \ell - 3/2 | n, \ell \rangle}} \end{aligned} \tag{4.24}$$

$$\boxed{|n-1, \ell+1\rangle = \frac{\zeta_\ell |n, \ell\rangle}{\sqrt{n-\ell}}} \tag{4.25}$$

Similarly

$$\boxed{|n+1, \ell\rangle = \frac{\zeta_\ell^\dagger |n, \ell+1\rangle}{\sqrt{n-\ell+1}}} \tag{4.26}$$

$$\boxed{|n-1, \ell\rangle = \frac{\xi_\ell |n, \ell+1\rangle}{\sqrt{n+\ell+2}}} \tag{4.27}$$

$$\boxed{|n+1, \ell+1\rangle = \frac{\xi_\ell^\dagger |n, \ell\rangle}{\sqrt{n+\ell+3}}} \tag{4.28}$$

We shall now calculate the wavefunctions $R_{n\ell}(r) \equiv \langle r | n, \ell \rangle$, begin by writing (4.21) in position representation

$$\Big(\frac{\partial}{\partial r} - \frac{n}{r} + \frac{\mu\omega r}{\hbar}\Big) R_{nn}(r) = 0 \tag{4.29}$$

This is a linear first order ODE, solved by separation of variables, the solution is

$$\boxed{R_{nn}(r) \propto r^n e^{-\mu\omega r^2/2\hbar}} \tag{4.30}$$

Every other radial wavefunction can be generated by using the appropriate ladder operators. Similarly to hydrogen, the full wavefunction is given by $\psi_{n\ell}^m(r, \theta, \phi) = R_{n\ell}(r) Y_\ell^m(\theta, \phi)$.

Unlike other problems encountered so far, there are 2 sets of ladder operators for the isotropic harmonic oscillator. Furthermore, these ladder operators changes two quantum numbers n and ℓ simutaneously.

5 Fock-Darwin Oscillator

The dynamics of the quantum dot can be well-described by the Fock-Darwin Hamiltonian, which is a 2D harmonic oscillator with a magnetic field applied perpendicular to the plane of oscillation. We begin by considering a 2D harmonic oscillator, then add the effect of magnetic field later. It will be shown to be convenient to write the Hamiltonian in polar coordinates:

$$H = \frac{p_r^2}{2\mu} - \frac{\hbar^2}{8\mu r^2} + \frac{L_z^2}{2\mu r^2} + \frac{1}{2}\mu\omega^2 r^2 \tag{5.1}$$

The fundamental commutator in this problem is analogous to the canonical commutation relation

$$[r, p_r] = i\hbar \tag{5.2}$$

As $[H, L_z] = 0$, there exist a complete set of mutual eigenstates $\{|n, m\rangle\}$

The effect of Hamiltonian H on mutual eigenstates is equivalent to the radial Hamiltonian defined as

$$H_m = \frac{1}{2\mu}\Big(p_r^2 + \frac{(4m^2-1)\hbar^2}{4r^2} + \mu^2\omega^2 r^2\Big) \tag{5.3}$$

We wish to solve the radial TISE

$$H_m|n, m\rangle = E_n|n, m\rangle \tag{5.4}$$

Consider the following operators

$$\boxed{\tau_m \equiv \frac{1}{\sqrt{2\mu\hbar\omega}}\Big(ip_r - \frac{(2m+1)\hbar}{2r} + \mu\omega r\Big)} \tag{5.5}$$

$$\boxed{\tau_m^\dagger = \frac{1}{\sqrt{2\mu\hbar\omega}}\Big(-ip_r - \frac{(2m+1)\hbar}{2r} + \mu\omega r\Big)} \tag{5.6}$$

$$\boxed{\pi_m \equiv \frac{1}{\sqrt{2\mu\hbar\omega}}\Big(ip_r + \frac{(2m+1)\hbar}{2r} + \mu\omega r\Big)} \tag{5.7}$$

$$\boxed{\pi_m^\dagger = \frac{1}{\sqrt{2\mu\hbar\omega}}\Big(-ip_r + \frac{(2m+1)\hbar}{2r} + \mu\omega r\Big)} \tag{5.8}$$

Which combines to give

$$\tau_m\tau_m^\dagger = \frac{H_{m+1}}{\hbar\omega} - m \tag{5.9}$$

$$\tau_m^\dagger\tau_m = \frac{H_m}{\hbar\omega} - m - 1 \tag{5.10}$$

$$\pi_m\pi_m^\dagger = \frac{H_m}{\hbar\omega} + m + 1 \tag{5.11}$$

$$\pi_m^\dagger \pi_m = \frac{H_{m+1}}{\hbar\omega} + m \tag{5.12}$$

We now demonstrate these are ladder operators, consider

$$\begin{aligned} H_{m+1}\tau_m|n,m\rangle &= \hbar\omega(\tau_m\tau_m^\dagger + m)\tau_m|n,m\rangle \\ &= \hbar\omega\tau_m(\tau_m^\dagger\tau_m + m)|n,m\rangle \\ &= \tau_m(H_m - \hbar\omega)|n,m\rangle \\ &= (E_n - \hbar\omega)\tau_m|n,m\rangle \end{aligned} \tag{5.13}$$

Comparison with radial TISE yields

$$\tau_m|n,m\rangle \propto |n-1,m+1\rangle \tag{5.14}$$

Following the same arguments yields

$$\tau_m^\dagger|n,m+1\rangle \propto |n+1,m\rangle \tag{5.15}$$

$$\pi_m|n,m+1\rangle \propto |n-1,m\rangle \tag{5.16}$$

$$\pi_m^\dagger|n,m\rangle \propto |n+1,m+1\rangle \tag{5.17}$$

Furthermore, examination of the effective potential suggests the ladder must terminate

$$\tau_n|n,n\rangle = 0 \tag{5.18}$$

where we have defined $n \equiv m_{max}$
We can then extract the eigenvalues

$$\begin{aligned} E_n &= \langle n,n|H_n|n,n\rangle \\ &= \hbar\omega\langle n,n|\tau_n^\dagger\tau_n + n + 1|n,n\rangle \end{aligned} \tag{5.19}$$

$$\boxed{E_n = (n+1)\hbar\omega} \tag{5.20}$$

Note the zero-point energy is $\hbar\omega/2$ in each dimension.
we shall now normalise the newly generate the mutual eigenkets in the following way

$$\begin{aligned} |n-1,m+1\rangle &= \frac{\tau_m|n,m\rangle}{\sqrt{\langle n,m|\tau_m^\dagger\tau_m|n,m\rangle}} \\ &= \frac{\tau_m|n,m\rangle}{\sqrt{\langle n,m|H_m/\hbar\omega - m - 1|n,m\rangle}} \end{aligned} \tag{5.21}$$

$$\boxed{|n-1,m+1\rangle = \frac{\tau_m|n,m\rangle}{\sqrt{n-m}}} \tag{5.22}$$

Similarly

$$\boxed{|n+1,m\rangle = \frac{\tau_m^\dagger |n,m+1\rangle}{\sqrt{n-m+1}}} \tag{5.23}$$

$$\boxed{|n-1,m\rangle = \frac{\pi_m |n,m+1\rangle}{\sqrt{n+m+1}}} \tag{5.24}$$

$$\boxed{|n+1,m+1\rangle = \frac{\pi_m^\dagger |n,m\rangle}{\sqrt{n+m+2}}} \tag{5.25}$$

We shall now calculate the radial wavefunctions $R_{nm}(r) \equiv \langle r|n,m\rangle$. Begin by writing (5.18) in position representation

$$\Big(\frac{\partial}{\partial r} - \frac{n}{r} + \frac{\mu\omega r}{\hbar}\Big) R_{nn}(r) = 0 \tag{5.26}$$

where we have used

$$p_r = -i\hbar\Big(\frac{\partial}{\partial r} + \frac{1}{2r}\Big) \tag{5.27}$$

(5.26) is a linear first order ODE, solved by separation of variables, the solution is

$$\boxed{R_{nn}(r) \propto r^n e^{-\mu\omega r^2/2\hbar}} \tag{5.28}$$

Every other radial wavefunction can be generated by using the appropriate ladder operators. The full wavefunction is given by

$$\psi_n^m(r,\phi) = R_{nm}(r) e^{im\phi} \tag{5.29}$$

We now introduce a new quantum number $u = 0, 1 \ldots$, this is the number of radial nodes. Further analysis shows:

$$\boxed{n = 2u + |m|} \tag{5.30}$$

When a magnetic field B is applied in the z direction, the Hamiltonian becomes

$$H_B = \frac{p_r^2}{2\mu} - \frac{\hbar^2}{8\mu r^2} + \frac{L_z^2}{2\mu r^2} + \frac{1}{2}\mu\omega_e^2 r^2 - \omega_B L_z \tag{5.31}$$

where $\omega_B \equiv eB/2\mu$ and $\omega_e \equiv \sqrt{\omega^2 + \omega_B^2}$. Replace L_z with its eigenvalue to obtain the radial Hamiltonian

$$H_{Bm} = \frac{1}{2\mu}\Big(p_r^2 + \frac{(4m^2-1)\hbar^2}{4r^2} + \mu^2\omega_e^2 r^2\Big) - m\hbar\omega_B \tag{5.32}$$

This is the original radial Hamiltonian with replacement $\omega \to \omega_e$ and an extra constant term added at the end, therefore the same set of ladder operators can be used if the replacement $\omega \to \omega_e$ is made.

The eigenvalues are

$$\boxed{E_{u,m} = (2u + |m| + 1)\hbar\omega_e - m\hbar\omega_B} \tag{5.33}$$

We choose to relabel the eigenstates $|n, m\rangle \to |u, m\rangle$ as n can no longer label the energy eigenvalue by itself.

The radial TISE becomes

$$H_{Bm}|u, m\rangle = E_{u,m}|n, m\rangle \tag{5.34}$$

Repeat the procedure (5.13) yields following results

$$H_{Bm+1}\tau_m|u, m\rangle = (E_{u,m} - \hbar\omega_e - \hbar\omega_B)\tau_m|u, m\rangle \tag{5.35}$$

$$H_{Bm}\tau_m^\dagger|u, m+1\rangle = (E_{u,m+1} + \hbar\omega_e + \hbar\omega_B)\tau_m^\dagger|u, m+1\rangle \tag{5.36}$$

$$H_{Bm}\pi_m|u, m+1\rangle = (E_{u,m+1} - \hbar\omega_e + \hbar\omega_B)\pi_m|u, m+1\rangle \tag{5.37}$$

$$H_{Bm+1}\pi_m^\dagger|u, m\rangle = (E_{u,m} + \hbar\omega_e - \hbar\omega_B)\pi_m^\dagger|u, m\rangle \tag{5.38}$$

Hence

$$\tau_m|u, m\rangle \propto |u-1, m+1\rangle \tag{5.39}$$

$$\tau_m^\dagger|u, m+1\rangle \propto |u+1, m\rangle \tag{5.40}$$

$$\pi_m|u, m+1\rangle \propto |u, m\rangle \tag{5.41}$$

$$\pi_m^\dagger|u, m\rangle \propto |u, m+1\rangle \tag{5.42}$$

Equation (5.39) to (5.42) is the same as (5.14) to (5.17) but in new notation. Hence, the structure of the $u - m$ ladder stays the same. These states can be normalised in the usual way. The form of the wavefunctions are the same, however, width of the Gaussian differs. Namely, a stronger magnetic field decreases the width of Guassian, thus better confining the electron on a circle.

Furthermore, the magnetic field removes the degeneracy of the eigenstates, which suggests the lowering of symmetry of the system.

6 Morse Oscillator

The Morse oscillator is a better approximation for the potential energy curve of a diatomic molecule than the harmonic potential, as it accounts for both anharmonicity and the dissociation of molecule. We shall solve its TISE in this section.

The Hamiltonian for a Morse oscillator is written as

$$H = \frac{p^2}{2\mu} + D_e(1 - e^{-ax})^2 \tag{6.1}$$

Where $D_e \equiv \hbar\omega/4\chi$ and $a \equiv \sqrt{2\mu\omega\chi/\hbar}$, $0 \leq \chi < 1$ is the anharmonicity constant of the oscillator.

Divide both sides of (6.1) by $\hbar\omega\chi$ yields

$$\frac{H}{\hbar\omega\chi} = \frac{p^2}{2\mu\hbar\omega\chi} + \frac{1}{4\chi^2}(1 - e^{-ax})^2 \tag{6.2}$$

We now define a crucial parameter for this problem:

$$\eta \equiv \frac{1}{2\chi} - \frac{1}{2} \tag{6.3}$$

Notice $\eta > 0$

Substitute (6.3) into (6.2) and rearrange yields

$$\frac{H}{\hbar\omega\chi} = \frac{p^2}{2\mu\hbar\omega\chi} + \left(\eta + \frac{1}{2} - \frac{1}{2\chi}e^{-ax}\right)^2 \tag{6.4}$$

H and its eigenstates are both labeled by η, we also define a dimensionless Hamiltonian H_η via the LHS of (6.4) to simplify algebra

$$H_\eta|n\rangle_\eta = E_{n,\eta}|n\rangle_\eta \tag{6.5}$$

Consider the following operators:

$$\boxed{\sigma_\eta \equiv \eta - \frac{1}{2\chi}e^{-ax} + \frac{ip}{\sqrt{2\mu\hbar\omega\chi}}} \tag{6.6}$$

$$\boxed{\sigma_\eta^\dagger = \eta - \frac{1}{2\chi}e^{-ax} - \frac{ip}{\sqrt{2\mu\hbar\omega\chi}}} \tag{6.7}$$

Which combines to give the Hamiltonian

$$\sigma_\eta^\dagger\sigma_\eta = H_\eta - \eta - \frac{1}{4} \tag{6.8}$$

$$\sigma_\eta\sigma_\eta^\dagger = H_{\eta-1} + \eta - \frac{1}{4} \tag{6.9}$$

We now demonstrate the proposed operators are indeed ladder operators.

Consider

$$
\begin{aligned}
H_\eta \sigma_\eta^\dagger |n\rangle_{\eta-1} &= \Big(\sigma_\eta^\dagger \sigma_\eta + \eta + \frac{1}{4}\Big)\sigma_\eta^\dagger |n\rangle_{\eta-1} \\
&= \sigma_\eta^\dagger \Big(\sigma_\eta \sigma_\eta^\dagger + \eta + \frac{1}{4}\Big)|n\rangle_{\eta-1} \\
&= \sigma_\eta^\dagger \Big(H_{\eta-1} + 2\eta\Big)|n\rangle_{\eta-1} \\
&= \Big(E_{n,\eta-1} + 2\eta\Big)\sigma_\eta^\dagger |n\rangle_{\eta-1}
\end{aligned} \tag{6.10}
$$

hence we can conclude

$$
\sigma_\eta^\dagger |n\rangle_{\eta-1} \propto |n+1\rangle_\eta \tag{6.11}
$$

$$
E_{n+1,\eta} = E_{n,\eta-1} + 2\eta \tag{6.12}
$$

Repeat the procedure (6.10) by considering $H_{\eta-1}\sigma_\eta |n\rangle_\eta$ yields

$$
\sigma_\eta |n\rangle_\eta \propto |n-1\rangle_{\eta-1} \tag{6.13}
$$

$$
E_{n-1,\eta-1} = E_{n,\eta} - 2\eta \tag{6.14}
$$

Notice that (6.14) is equivalent to (6.12) with substitution $n \to n+1$.

H_η is a sum of squares of Hermitian operators, thus its eigenvalue cannot be negative. Thus there exist a ground state which satisfies the following relation

$$
\sigma_\eta |0\rangle_\eta = 0 \tag{6.15}
$$

Thus the zero-point energy (divided by $\hbar\omega\chi$) of a Morse oscillator is

$$
E_{0,\eta} = \eta + \frac{1}{4} \tag{6.16}
$$

The rest of the energy spectrum is extracted by repeated substitution of recurrence relation (6.14) into itself

$$
\begin{aligned}
E_{n,\eta} &= E_{n-1,\eta-1} + 2\eta \\
&= E_{n-2,\eta-2} + 2\eta + 2(\eta-1) \\
&= E_{0,\eta-n} + 2\eta + 2(\eta-1) + 2(\eta-2) + \cdots + 2(\eta-n+1) \\
&= \eta - n + \frac{1}{4} + \sum_{k=0}^{n-1} 2(\eta-k) \\
&= \eta - n + \frac{1}{4} + 2n\eta - 2\sum_{k=0}^{n-1} k \\
&= \eta - n + \frac{1}{4} + 2n\eta - n(n-1) \\
&= (2n+1)\eta - n^2 + \frac{1}{4}
\end{aligned} \tag{6.17}
$$

Substitute (6.3) then multiply by $\hbar\omega\chi$ we divided through at the start and after a slight rearrangement yields

$$\boxed{E_n = \left(n + \frac{1}{2}\right)\hbar\omega - \left(n + \frac{1}{2}\right)^2 \hbar\omega\chi} \tag{6.18}$$

The following relation can be deduced by repeated substitution of recurrence relation (6.11) into itself

$$\boxed{|n\rangle_\eta \propto \sigma_\eta^\dagger \sigma_{\eta-1}^\dagger \cdots \sigma_{\eta-n+1}^\dagger |0\rangle_{\eta-n}} \tag{6.19}$$

Where a total of n $\sigma^\dagger$s are used.

Notice $\eta' = \eta - n$ must be positive, thus we deduce

$$\boxed{\lfloor \eta \rfloor = n_{max}} \tag{6.20}$$

Thus there is exactly $n_{max} + 1$ bound states.

We shall now normalise the eigenstates

$$|n+1\rangle_\eta = \frac{\sigma_\eta^\dagger |n\rangle_{\eta-1}}{\sqrt{{}_{\eta-1}\langle n|\sigma_\eta \sigma_\eta^\dagger |n\rangle_{\eta-1}}} \tag{6.21}$$

Using (6.9) and (6.17) yields

$$\boxed{|n+1\rangle_\eta = \frac{\sigma_\eta^\dagger |n\rangle_{\eta-1}}{\sqrt{(n+1)(2\eta - (n+1))}}} \tag{6.22}$$

Similarly

$$\boxed{|n-1\rangle_\eta = \frac{\sigma_{\eta+1} |n\rangle_{\eta+1}}{\sqrt{n(2(\eta+1) - n)}}} \tag{6.23}$$

We now calculate the wavefunctions $\langle x|n\rangle$

We begin by casting (6.15) into the position representation

$$\left(\eta - \frac{1}{2\chi} e^{-ax} + \sqrt{\frac{\hbar}{2\mu\omega\chi}} \frac{d}{dx}\right) \langle x|0\rangle_\eta = 0 \tag{6.24}$$

This is a first order linear ODE, which can be solved easily by separation of variables, the solution is

$$\boxed{\langle x|0\rangle_\eta \propto e^{-e^{-ax}/2\chi - \eta a x}} \tag{6.25}$$

Every other wavefunction can be generated by using the relation (6.19).

Furthermore, we notice that $\langle x|0\rangle_\eta$ diverges, thus cannot be normalised if $\eta \leq 0$, this is consistent with the definition of η.

In this note, we have solved six eigenvalue problems via finding the ladder operators. These ladder operators allow us to extract the eigenvalue and eigenstates with relative ease compared to solving a difficult second-order differential equation by going into the position representation from the outset. This entire note can be regarded as a demonstration of the elegance of the intrinsic algebraic structure of quantum mechanics.